Response to reviewer comments on "Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and the Bayesian evidence in early-phase clinical trials"

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Response to Associate Editor's Comments

The paper introduces a generalized form of Jeffreys's approximate objective Bayes factor (eJAB). It was read by myself and two expert referees. The authors establish consistency and assess finite-sample accuracy via simulations. There were numerous major concerns identified by the reviewers:

1. *'more exploration of the connection to test-based BF (Johnson, 2005)'*

   We originally selected test-statistic Bayes factors (TSBFs) for two main reasons: their computational convenience, as they admit closed-form expressions (although TSBFs are not 'approximate' Bayes factors in the same sense as eJAB), and the lack of readily available MCMC-based Bayes factors for chi-squared and other nonparametric tests considered in our simulations. In our response to Reviewer 1's comment, we now discuss the assumptions and asymptotic properties of three versions of the TSBF and compare them with those of eJAB. We have also revised the simulation studies to avoid comparisons between eJAB and the minimum Bayes factor version of TSBF.

   The Equally Weighted Decision Error Profiles compare eJAB with TSBFs using true and estimated effects, NHST significance rules, and BIC approximations. The revised manuscript includes BIC as Panel D of Figure 2, and its abstract now summarizes these comparisons and the dependence of relative performance on sample size, effect size, and dimension.

2. *'significant concerns about the regularity condition R2'*

   In our response to Reviewer 1, we have carefully addressed concerns regarding the regularity condition R2 required for model-selection consistency. First, we have discussed asymptotic properties of TSBF in relation to R2 in our first response to Reviewer 1. In our second response to Reviewer 1, we have made comparisons with the BIC approximation to the Bayes factor and the conditions required for model selection consistency in relation R2. Finally, we have conducted an empirical investigation of R2 based on data obtained from four large scale replication studies, where we observe realizations of D_n under H_1 and comment on the behavior of D_n with increasing sample size.

3. *'significant concerns about the analysis in Section 4.'*

   We have revised the analysis in response to Reviewer 1's suggestions. The application is now restricted to 16,563 primary- and secondary-outcome analyses from 1,790 early-phase trials, with detailed inclusion and exclusion criteria provided in Appendix C. One-sided tests are now considered explicitly in addition to two-sided tests. The revised analysis is retrospective and exploratory; it was neither prospectively preregistered nor independently replicated in a second database.

4. *Addition of new coauthor:* We have added one new coauthor to the manuscript. Evan Strasdin is a graduate student who conducted the empirical study that forms the second part of our response to the second comment of Reviewer 1.

5. *Minor modification to the title of the manuscript:* We have changed the title of the manuscript from '*Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and statistical evidence in 71,126 clinical trial findings*' to '*Generalized Jeffreys's approximate objective Bayes factor: Model-selection consistency, finite-sample accuracy, and the Bayesian evidence in early-phase clinical trials*'. This change is required as part of our response to the fifth comment of Reviewer 1, who suggested that we only look at early phase trials (phase I and II) in our analysis.

Response to Reviewer 1

- *The proposed BF depends on the test-statistic of a chi-squared test with dimension q. It also depends on sample size n. The connection to the test-based BF (Johnson, 2005) should be explored in more detail, both theoretically and through simulation. For example, Johnson also considers BFs based on the F- (rather than the chi-squared) statistic, which also depend on sample size. In fact, you do use F-tests in simulation items 4 and 5. A closer theoretical comparison with BFs based on a chi-squared test statistic is warranted. The derivation in Johnson is based on certain asymptotic assumptions (local alternatives); how do these compare with assumptions R1 and, in particular, R2?*

  Thank you for raising this question. We have expanded the discussion to distinguish **three** related constructions: the local alternative prior, the maximum marginal likelihood estimate (MMLE), and the nonlocal alternative prior. Johnson's TSBF models the sampling distribution of a classical test statistic rather than the full-data likelihood. For the chi-squared test, the statistic has a central chi-squared distribution under $\mathcal{H}_0$ and a noncentral chi-squared distribution under $\mathcal{H}_1$, yielding a closed-form 'Bayes factor' that depends on the statistic x_n, its degrees of freedom q, the sample size n, and an alternative-model dispersion parameter $\varphi = (c+1)/n$, where $c = \mathcal{O}(n)$ denotes the Dirichlet concentration parameter,

$$\mathcal{H}_0 : \; x_n \sim \chi^2_q, \tag{1}$$

$$\mathcal{H}_1 : \; x_n \mid \gamma \sim \chi^2_q(\gamma), \qquad \gamma \mid \varphi \sim \text{Gamma}\left(\frac{q}{2}, \frac{\varphi}{2}\right), \tag{2}$$

$$\mathit{TSBF}_{10} = \left(\frac{\varphi}{\varphi+1}\right)^{\frac{q}{2}} \exp\left\{\frac{x_n}{2(\varphi+1)}\right\}. \tag{3}$$

  Johnson (2005) considers maximizing the marginal density of the observed test statistic under $\mathcal{H}_1$ over the dispersion parameter. This produces the minimum BF_{01} within the specified class (Held and Ott, 2018). It is therefore an upper bound and the least conservative assessment of evidence against $\mathcal{H}_0$. Our previous manuscript compared eJAB only with this MMLE version, rather than with the broader class of TSBFs (Johnson, 2005, 2008; Yuan and Johnson, 2008; Johnson, 2013). Because this quantity is maximized in favor of $\mathcal{H}_1$, it is bounded below by one and equals one when the observed statistic does not exceed its expectation under $\mathcal{H}_0$, producing the characteristic 'hockey-stick' behavior. We agree that comparing an approximate Bayes factor with a minimum Bayes factor is an apples to oranges comparison. Accordingly, we have removed the comparisons with the MMLE-TSBF and replaced them with comparisons to the TSBF more generally.

  The asymptotic assumptions differ in their intent. To test the hypotheses $\mathcal{H}_0 : \boldsymbol{\theta} = \boldsymbol{\theta}_0$ versus $\mathcal{H}_1 : \boldsymbol{\theta} \neq \boldsymbol{\theta}_0$, Johnson's original chi-squared derivation assumes local alternatives satisfying $\boldsymbol{\theta} - \boldsymbol{\theta}_0 = \mathcal{O}_p(n^{-\frac{1}{2}})$. In other words, a local alternative not only centers a unimodal, symmetric prior distribution at the null value but also specifies that the prior scale shrinks with the sample size. One consequence of this assumption is the existence of a nondegenerate asymptotic distribution for the test statistic under $\mathcal{H}_1$—specifically, a noncentral chi-squared limit with an $\mathcal{O}_p(1)$ noncentrality parameter (Johnson, 2005, Lemma 1). It is not a condition that ensures model-selection consistency. Under this local-alternative regime, regularity condition R2 will not hold, and model-selection consistency of eJAB is not guaranteed. However, we note that the TSBF is also not model-selection consistent under the same assumption. Under $\mathcal{H}_0$, $x_n \xrightarrow{D} \chi^2_q$ and $\varphi \to \varphi_0 > 0$ as $n \to \infty$, so TSBF_{10} in Eq. 3 is nondegenerate and almost surely positive. Thus, under Johnson's local-alternative framework, the asymptotic TSBF exists in a nondegenerate form, but neither eJAB nor the TSBF is model-selection consistent. By contrast, eJAB is based on three asymptotic assumptions: (1) Asymptotic normality of the posterior distribution; (2) R1: $p \xrightarrow{D} \text{Unif}(0,1)$ as $n \to \infty$ under $\mathcal{H}_0$; (3) R2: $D_n = -\sqrt{n} \cdot p \ln p \xrightarrow{P} 0$ as $n \to \infty$ under $\mathcal{H}_1$. The first assumption along with the assumed prior gives us the form for eJAB. The latter two assumptions are sufficient for model-selection consistency (i.e., Theorem 2.2). Under these assumptions, the same argument shows that TSBF_{10} in Eq. 3 does not converge to zero under $\mathcal{H}_0$ and remains

model-selection inconsistent. We have revised the manuscript to clarify the differences in assumptions and their implications.

We further consider the model selection consistency of TSBF in subsequent work, across three recent papers, the Bayes factor function (BFF) retains the same core structure—a family of TSBFs evaluated over alternative hypotheses indexed by standardized effect size ω—while the principal extensions concern the nonlocal priors used to define those alternatives, expanding the first-order moment and gamma priors to arbitrary-order moment and inverse-moment/inverse-gamma families and examining how their shape and dispersion affect evidence accumulation, operating characteristics, and aggregation across replicated studies (Johnson et al., 2023; Datta et al., 2025a,b). The following expressions give the BFF for the chi-squared test proposed by Johnson et al. (2023),

$$\mathcal{H}_0 : \; x_n \sim \chi^2_q, \tag{1}$$

$$\mathcal{H}_1 : \; x_n \mid \gamma \sim \chi^2_q(\gamma), \qquad \gamma \mid \tau^2 \sim \text{Gamma}\left(\frac{q}{2}+1, \, \frac{1}{2\tau^2}\right), \tag{4}$$

$$BFF_{10}\left(x_n \mid \tau^2\right) = (1+\tau^2)^{-\frac{q}{2}-1}\left(1+\frac{\tau^2 x_n}{q(1+\tau^2)}\right)\exp\left\{\frac{\tau^2 x_n}{2(1+\tau^2)}\right\}, \tag{5}$$

$$\tau^2 = \frac{n}{q}\omega^2. \tag{6}$$

They assume the alternative in Eq. 4 so that the prior density on the noncentrality parameter vanishes at the origin. Since $\text{mode}(\gamma) = q\tau^2$, Eq. 6 matches the prior mode to the noncentrality $n\omega^2$ corresponding to a specified standardized effect size ω. This first-order nonlocal prior yields faster accumulation of evidence in favor of a true null than the corresponding local prior. Under $\mathcal{H}_0$, $x_n = \mathcal{O}_p(1)$ and, for every fixed $\omega > 0$, $\tau^2 = \mathcal{O}(n)$, so $BFF_{10} = \mathcal{O}_p\left(n^{-\frac{q}{2}-1}\right) \xrightarrow{P} 0$ as $n \to \infty$. The faster polynomial accumulation of null evidence motivates the higher-order nonlocal priors considered in Datta et al. (2025a). Under a fixed alternative with true standardized effect $\omega_0 > 0$, the true noncentrality parameter satisfies $\gamma = n\omega_0^2$, and hence $x_n = n\omega_0^2 + \mathcal{O}_p(\sqrt{n})$. For any fixed nonzero BFF ordinate $\omega > 0$, $\tau^2 = \mathcal{O}(n)$, and therefore the exponential term in Eq. 5 dominates its polynomial penalty. In particular, $\frac{1}{n}\ln BFF_{10} \xrightarrow{P} \frac{\omega_0^2}{2}$, so $BFF_{10} \xrightarrow{P} \infty$ as $n \to \infty$. Thus, for every fixed nonzero effect-size ordinate ω, the BFF is pointwise model-selection consistent.

- *More theoretical motivation and discussion are needed for regularity condition R2. It is a stronger condition than assuming that the traditional minimum BF calibration, $-e \cdot p \ln p$ (Sellke et al., 2001), converges to zero under $\mathcal{H}_1$ as n goes to infinity. So how strong is condition 2? Is this assumption justifiable?*

Assume that, under $\mathcal{H}_1$, the p-value is bounded away from 1 for all sufficiently large n. Then, $-e \cdot p \ln p \xrightarrow{P} 0$ is equivalent to $p \xrightarrow{P} 0$ because $\lim_{x \to 0^+} x \ln x = 0$. Thus, requiring the traditional minimum-BF calibration to converge to zero imposes no restriction on the rate at which the p-value converges to zero under $\mathcal{H}_1$. With regard to the strength of assumption R2: $D_n = -\sqrt{n} \cdot p \ln p \xrightarrow{P} 0$ as $n \to \infty$ under $\mathcal{H}_1$, we first note that it is not sufficient to ensure model-selection consistency of the BIC approximation to the Bayes factor, as formulated in Eq. 7. Let $Q_{\chi^2_q}(1-p)$ denote the quantile function of a chi-squared distribution with q degrees of freedom, evaluated at tail probability $1-p$, where $p \to 0^+$. Suppose that the p-value is computed from a likelihood-ratio test so that, under standard regularity conditions,

$$BF_{01}^{(\text{BIC})} = n^{\frac{q}{2}} \exp\left\{-\frac{1}{2}Q_{\chi^2_q}(1-p)\right\}, \tag{7}$$

and that R2 holds, but $-n^a \cdot p \ln p \xrightarrow{P} 0$ for $a > \frac{1}{2}$ as $n \to \infty$ under $\mathcal{H}_1$. Then, $\exists \epsilon > 0$ such that $\mathbb{P}(-n^a \cdot p \ln p < \epsilon) \nrightarrow 1$, which, in turn, implies that $\mathbb{P}(BF_{01}^{(\text{BIC})} < \epsilon) \nrightarrow 1$ for $q > 1$ under $\mathcal{H}_1$.

More specifically, using the asymptotic expansion of the incomplete gamma function (e.g., Daouia et al., 2025, Lemma A.2), $\frac{1}{2}Q_{\chi^2_q}(1-p) = -\ln p + (\frac{q}{2}-1)\ln(-\ln p) - \ln\Gamma(\frac{q}{2}) + o(1)$. It follows that $BF_{01}^{(\mathrm{BIC})} \approx \Gamma(\frac{q}{2})\, n^{\frac{q}{2}} \cdot p\,(-\ln p)^{1-\frac{q}{2}}$, which requires a substantially stronger condition than R2 for model-selection consistency when $q > 1$.

Second, considering R2 from a practical standpoint, we examine the realized values of D_n in four large-scale replication projects. Each project selected a collection of previously published findings and attempted to replicate them under a preregistered protocol, so that each original result was paired with an independently conducted replication of the same effect.

- **EERP**, the Experimental Economics Replication Project (Camerer et al., 2016), which attempted to replicate laboratory experiments published in the *American Economic Review* and the *Quarterly Journal of Economics* between 2011 and 2014;
- **RPP**, the Reproducibility Project: Psychology (Open Science Collaboration, 2015), which attempted to replicate a collection of findings drawn from three high-impact psychology journals;
- **RPCB**, the Reproducibility Project: Cancer Biology (Errington et al., 2014), which considered experiments reported in high-impact preclinical cancer biology papers; we use the project's final analysis (Errington et al., 2021); and
- **SCORE**, Systematizing Confidence in Open Research and Evidence (Tyner et al., 2025), which assembled and considered replication of a large corpus of published claims from the social and behavioral sciences.

For each reported result in each project, we extracted the original study's p-value and sample size, together with the degrees of freedom q for the corresponding test. We retained a result only when all three quantities were available and admissible. Table 1 reports the resulting counts for which we have original results and results of replication attempts. We then restrict attention to those results for which the original p-value satisfies $p \leq 0.05$ and the effect was successfully replicated. Given that the original result is replicated, we make the assumption that $\mathcal{H}_1$ is true in such cases.

		Results		$\mathcal{H}_1$ subset	
Project	Field	reported	extracted	original	replication
EERP	Experimental economics	18	18	10	10
RPP	Psychology	167	94	33	33
RPCB	Preclinical cancer biology	187	112	50	61
SCORE	Social and behavioral sciences	287	287	174	199
	Total	659	511	267	303

Table 1: Results extracted from each replication project and sizes of the $\mathcal{H}_1$ subset used in Figure 1. 'Reported' denotes the number of results released by each project, whereas 'extracted' denotes the number retained after applying the admissibility criteria ($0 < p < 1$, $n > 1$, and $q \geq 1$). The replication counts exceed the original counts for RPCB and SCORE because, in those projects, some claims were replicated independently by more than one team, and each attempt contributes its own realized value. Working against this, five SCORE claims contribute no replication value at all, their replication p-value having been reported as zero and not being recoverable from the reported test statistics.

Figure 1 plots the realized values of D_n against the sample size n, which spans roughly seven orders of magnitude. Thus, for effects of the magnitude represented in these corpora, R2 appears feasible.

- *The analysis described in Section 4 is unconvincing for several reasons. First, clinical trial findings are usually based on one-sided tests at significance level* $\alpha = 0.025$.

In the revised analysis, we determine the test direction from the reported statistical method. For explicitly one-sided scalar tests ($q = 1$), we apply the one-sided eJAB discussed below and use $p_1 < .025$;

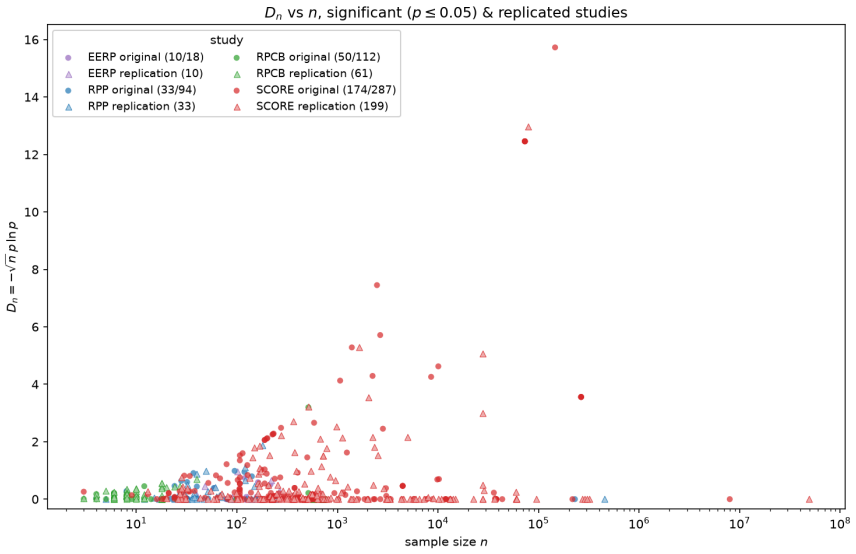


Figure 1: Realized values of $D_n = -\sqrt{n} \cdot p \ln p$ plotted against sample size n for the $\mathcal{H}_1$ subsets of the four replication projects (original $p \leq 0.05$ and successful replication). Circles denote the original studies and triangles their replications, with symbols colored by project. Legend entries give the size of each $\mathcal{H}_1$ subset relative to the number of results extracted from the corresponding project.

for two-sided or otherwise unrestricted tests, we apply the standard eJAB and use $p < .05$. This classification yields 427 one-sided and 16,136 two-sided or unrestricted analyses. When the test direction is not reported, we assign the analysis to the latter category. Appendix C.5 provides the classification rules and describes the treatment of reported inequalities.

- *The proposed BF, however, is intrinsically two-sided, as it is based on a chi-squared statistic. Can you amend eJAB to the one-sided setting, taking into account the direction of the effect? A one-sided Bayes factor may need different priors and assumptions from those used here.*

We have extended eJAB to the one-sided setting, e.g., $\mathcal{H}_0 : \theta = \theta_0$ versus $\mathcal{H}_+ : \theta > \theta_0$, and added the corresponding discussion, which explicitly incorporates the direction of the effect and the assumptions required for the one-sided Bayes factor. We have provided Appendix C.5 accordingly.

The encompassing-prior correction of Wagenmakers (2022, pp. 20–22) multiplies an unrestricted Bayes factor by the prior-to-posterior probability ratio for the predicted direction. For a scalar right-sided test, we use a symmetric local prior centered at θ_0, with prior mass $1/2$ in that direction and approximate posterior mass $1 - p_1$. With $p = 2\min\{p_1, 1 - p_1\}$, the null-centered Savage–Dickey approximation and the directional correction give

$$eJAB_{0+} = \frac{\sqrt{n}}{2(1 - p_1)} \exp\left\{-\frac{1}{2} Q_{\chi^2_1}(1 - p)\right\}.$$

Our finite-sample convention omits $(n-1)/n$ from the exponent of this null-centered approximation; the directional correction is the separate factor $1/[2(1 - p_1)]$. The two-sided eJAB in Definition 2.1 retains $(n - 1)/n$ for $q = 1$ and uses the MLE-centered prior. The posterior-mass approximation assumes a moderate sample size and a relatively vague local prior.

To establish one-sided consistency under this convention, write $x = Q_{\chi^2_1}(1 - p)$. The exact relation

between the two definitions is

$$eJAB_{0+} = \frac{eJAB_{01}\exp\{-x/(2n)\}}{2(1-p_1)}.$$

Under $\mathcal{H}_0$, R1 gives $x = \mathcal{O}_p(1)$, so $\exp\{-x/(2n)\} \xrightarrow{P} 1$. Since $[2(1-p_1)]^{-1} \geqslant 1/2$, Theorem 2.2 implies $eJAB_{0+} \xrightarrow{P} \infty$. Under $\mathcal{H}_+$, $\exp\{-x/(2n)\} \leqslant 1$, so R2 and the additional condition R3: $(1-p_1)^{-1} = \mathcal{O}_p(1)$ imply $eJAB_{0+} \xrightarrow{P} 0$. This retains the finite-sample factor when relating the one-sided and two-sided definitions.

- *Second, if you are concerned that 'failures at later phases may reflect type I errors, where the null hypothesis was incorrectly rejected at an earlier phase', then you should investigate only early-phase studies (Phases I and II) in your analysis. However, the analysis includes Phase II and even Phase IV studies.*

  To address this concern, we now restrict the manuscript primary- and secondary-outcome analysis to 16,563 eligible analyses from 1,790 exact Early Phase I, Phase I, and Phase II interventional trials. Combined-phase trials and Phase III and IV trials are excluded. We report the contradiction-rate profiles and posterior densities separately for the three classifications so that Early Phase I and Phase I are not combined. The observed contradiction proportions among direction-specifically significant results are 5/90 (5.56%), 25/918 (2.72%), and 431/5,401 (7.98%), respectively. These differ from the proportions among all eligible analyses in Figure 2 and the posterior summaries in Figure 3.

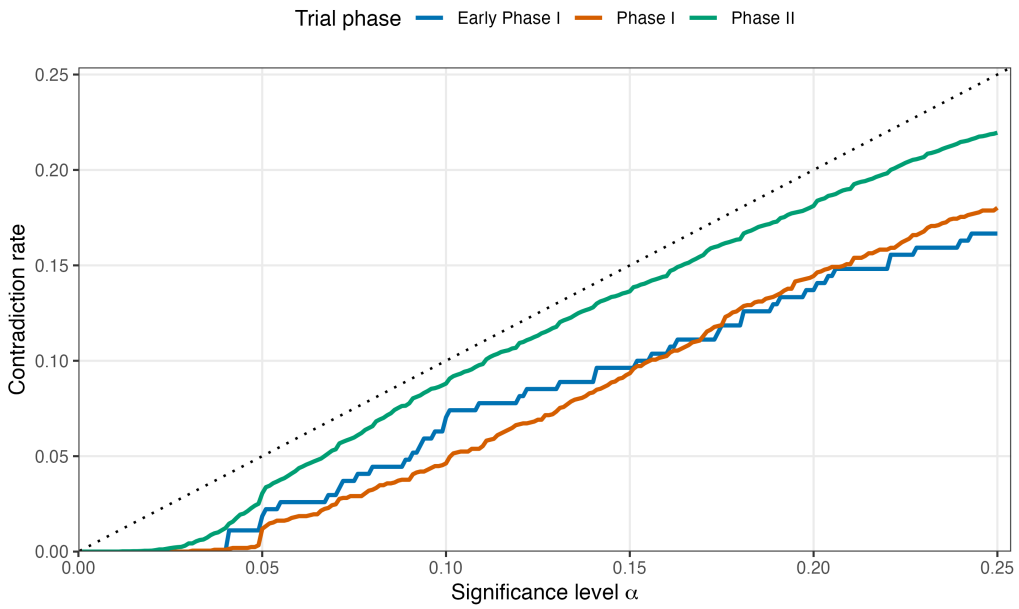


Figure 2: Bayes/NHST contradiction rate versus significance level among all eligible analyses in each trial phase. The sweep uses p for two-sided/unrestricted tests and $2\min\{p_1, 1-p_1\}$ for one-sided tests. The latter also admits opposite-tail results and differs from the directional rule used for the reported counts.

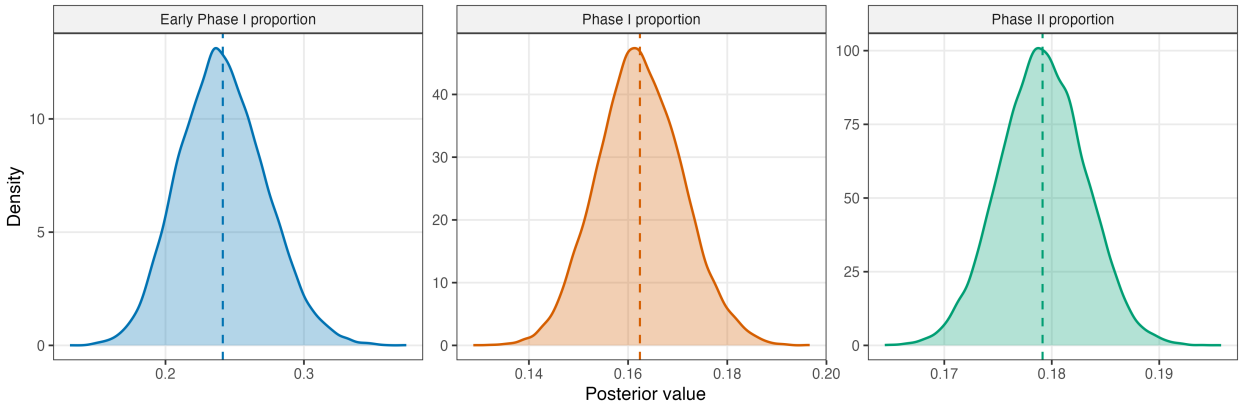


Figure 3: Study-level posterior distributions of the contradiction proportion among direction-specifically significant results, shown separately for the three trial phases. Dashed lines mark posterior means. The Dirichlet$(1, 1, 1, 1)$ prior applied to each study affects these estimates, which differ from the raw observed proportions.

- *Third, to justify such broad claims about type I errors in clinical trial findings, this study should be much more rigid, with a predefined research question and a study protocol appropriately preregistered before the analysis is conducted. But the study is purely exploratory, and it remains unclear how the 71,126 CTGs were selected, why the described analysis pipeline ('We fit a ... model', etc.) is appropriate, and so on. On the positive side, I appreciate the availability of code on OSF.*

  Our analysis was not prospectively preregistered, and we therefore make more limited claims about Bayes/NHST contradictions. The August 2025 CTG snapshot had already been downloaded and inspected before the July 2026 protocol was archived. Our aim is for this retrospective study to generate hypotheses, and we therefore consider an exploratory analysis appropriate. Contradictions do not establish that the null hypothesis is true, and their relationship with α does not estimate the type I error rate.

  We provide detailed inclusion and exclusion criteria for the clinical trials in Appendix C, together with the sequential flow from 279,927 analyses in 22,454 studies to the final 16,563 analyses in 1,790 trials. Both significant and nonsignificant eligible results are retained. The appendix specifies the test mappings, sample-size and dimension rules, and treatment of reported inequalities. For left-censored values such as $p < .05$, we retain the inequality for significance classification and use the numerical bound for eJAB. This can overstate evidence for the null, so the resulting contradiction classifications are provisional.

- *Section 2: 'More generally, the volume of prior probability ellipsoids around the MLE is $\mathcal{O}(n^{\frac{1-q}{2}})$.' So what? Why is this particular prior choice being made? Is this good or bad? Without appropriate justification, the proposed prior and BF are (for $q > 1$) ad-hoc.*

  The empirically derived prior distribution underlying eJAB is

$$\boldsymbol{\theta} \sim \boldsymbol{\mathcal{N}}_q\left(\hat{\boldsymbol{\theta}},\ n^{\frac{1}{q}} \cdot \mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})\right), \tag{8}$$

  where $\mathbf{J}_n = \mathcal{O}(n)$ denotes the $q \times q$ information matrix. The volume of a fixed-probability ellipsoid is proportional to $|n^{\frac{1}{q}} \cdot \mathbf{J}_n^{-1}|^{\frac{1}{2}} = \mathcal{O}(n^{\frac{1-q}{2}})$. More generally, suppose that $\boldsymbol{\theta} \sim \boldsymbol{\mathcal{N}}_q\left(\hat{\boldsymbol{\theta}},\ a_n \cdot \mathbf{J}_n^{-1}(\hat{\boldsymbol{\theta}})\right)$,

with $a_n > 1$. This yields $eJAB^*_{01} = a_n^{\frac{q}{2}} \exp\left\{-\frac{1}{2}\left(1 - \frac{1}{a_n}\right) Q_{\chi^2_q}(1-p)\right\}$ under the same Savage–Dickey approximation. Setting $a_n = n^{\frac{1}{q}}$ yields $\sqrt{n}$ growth in the evidence when $\mathcal{H}_0$ is true, so that under eJAB the rate of evidence accumulation towards $\mathcal{H}_0$ is independent of the dimension q. When $q = 1$, Eq. 8 reduces to the unit-information prior used in the original JAB (Aust et al., 2026; Wagenmakers, 2022). When $q > 1$ (F- and chi-squared tests) and effect sizes are small, it avoids the overly conservative behavior of the BIC approximation to the Bayes factor in favor of the null hypothesis (as detailed in the `R` Markdown files in the GitHub Simulations directory), while retaining model-selection consistency under R1 and R2.

Equally Weighted Decision Error Profiles: For each effect size θ and sample size n, we compare a simple null (effect zero) with a simple alternative (effect θ). Under 0–1 loss, the equally weighted decision error (EWDE) for decision rule d is

$$\mathrm{EWDE}_d(\theta, n) = \frac{\alpha_d(\theta, n) + \beta_d(\theta, n)}{2},$$

where α_d is the type I error probability under the null and β_d is the type II error probability at the specified alternative, equal to one minus power. We define $\Delta E = \mathrm{EWDE}_{\mathrm{comparison}} - \mathrm{EWDE}_{\mathrm{eJAB}}$ and plot the normalized EWDE comparison,

$$\mathrm{EWDE}_{\mathrm{norm}}(\theta, n) = \frac{\Delta E(\theta, n)}{M}, \qquad M = \max_{(\theta,n)\in\mathcal{G}} |\Delta E(\theta, n)|,$$

where $\mathcal{G}$ is the displayed grid for each panel. Positive values favor eJAB; negative values favor the comparison rule. This normalized EWDE comparison summarizes relative decision error while accounting for both types of error.

Figure 2 of the revised manuscript now combines estimated-effect TSBF (A), true-effect TSBF (B), $p < .05$ (C), and BIC (D). The BIC comparison uses $BF^{\mathrm{BIC}}_{01} = n^{q/2}\exp(-\Lambda/2)$, where Λ is the fitted likelihood-ratio statistic, for the six parametric-likelihood tests with available comparisons; Cox regression and the three rank tests are excluded. The figures below also retain the $p < .005$ comparison. The manuscript abstract and discussion now identify the observed regimes. For Pearson chi-squared independence tests in 2×2 and 3×3 tables, eJAB generally improves on estimated-ω TSBF outside small-sample or weak-signal regimes; small 3×3 samples can favor TSBF even at stronger effects. The pattern broadly reverses for 5×5 tables. With true ω, eJAB generally matches or underperforms TSBF, with differences that are not always small near weak signals.

Against BIC, single-parameter comparisons are close. The largest eJAB gains occur at weaker effects in ANOVA, repeated-measures ANOVA and chi-squared independence tests; at stronger effects, BIC can gain through lower type I error once both rules have high power. Against the NHST significance rule $p < .05$, eJAB generally gains with larger samples and stronger signals, whereas weak signals and some small-sample settings favor NHST. These are descriptions of the existing grids, not claims of uniform dominance.

- *Section 3: A table with the different designs (the ten tests), with an additional column for the degree of freedom, would be helpful. Then the second paragraph in Section 3.1 can be shortened.*

  We have added degrees of freedom to the panel facets and removed the second paragraph.

- *Section 3: 'effective sample size for survival analysis': A reference to Volinsky and Raftery (2000) might be useful. Also, it is not clear to me how the survival and censoring times were sampled.*

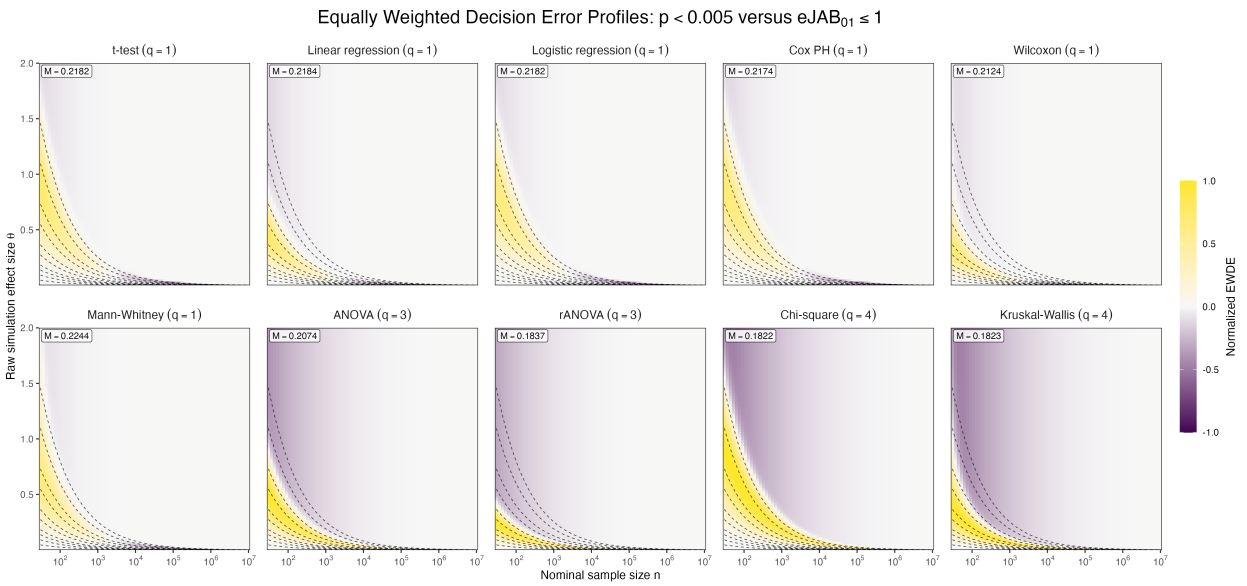


Figure 4: Equally Weighted Decision Error Profiles comparing eJAB with $p < .005$ across ten tests. The eJAB decision threshold is $BF_{01} \leq 1$. Colors show normalized EWDE; positive values favor eJAB and negative values favor the NHST significance rule.

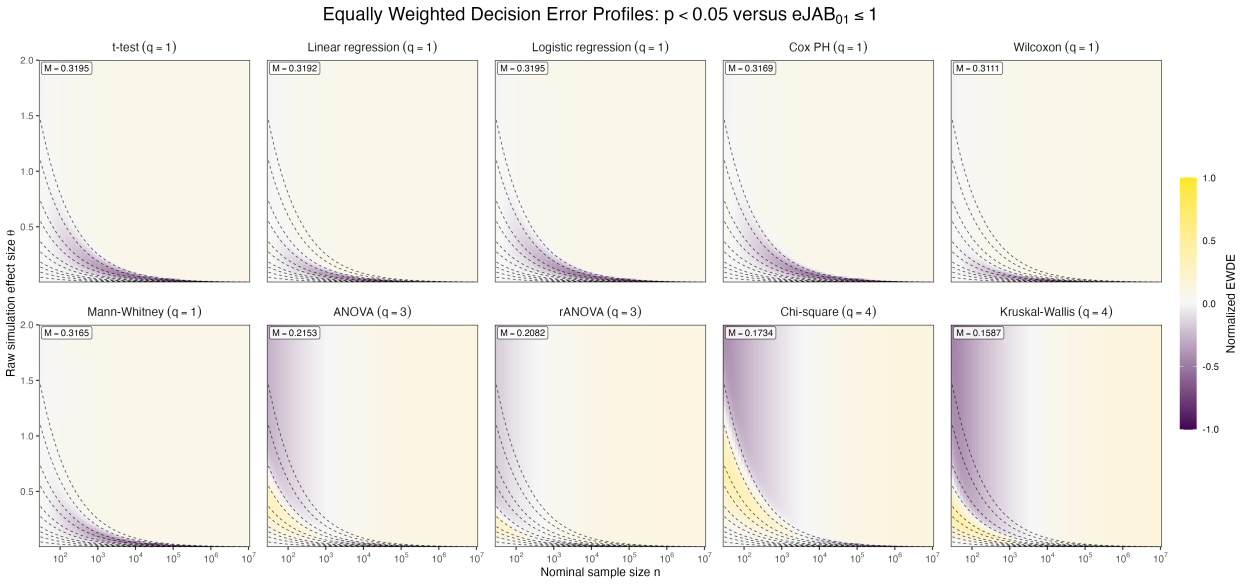


Figure 5: Equally Weighted Decision Error Profiles comparing eJAB with $p < .05$ across ten tests. The eJAB decision threshold is $BF_{01} \leq 1$. Colors show normalized EWDE; positive values favor eJAB and negative values favor the NHST significance rule.

We have added a reference to Volinsky and Raftery (2000) for the event-count definition of effective sample size. The survival designs discussed here are simulation studies. Survival analyses are excluded from the revised CTG application when the required event counts are unavailable, as documented in Appendix C.

In the simulations, survival times and censoring times are generated independently for each subject. The treatment indicator is binary, with approximately equal numbers of subjects assigned to the two

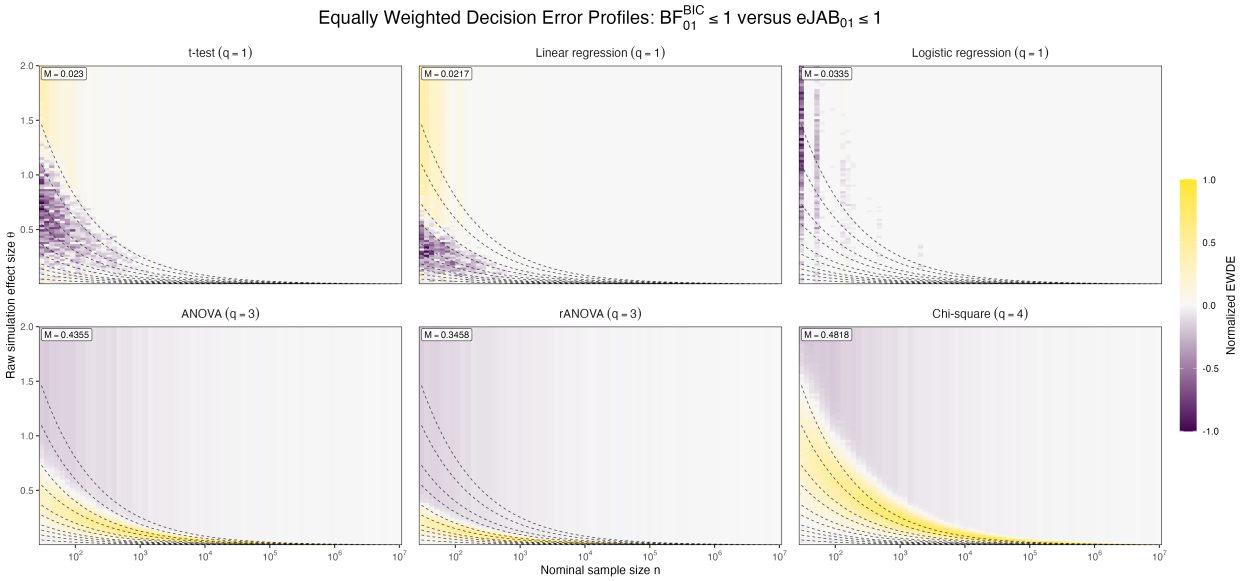


Figure 6: Equally Weighted Decision Error Profiles comparing eJAB with the BIC Bayes-factor rule across six parametric-likelihood tests: the two-sample t-test, linear regression, logistic regression, ANOVA, repeated-measures ANOVA, and chi-squared tests. Both Bayes-factor decision thresholds are $BF_{01} \leq 1$. Colors show normalized EWDE; positive values favor eJAB and negative values favor BIC. Each panel reports its normalization constant M, the maximum absolute EWDE difference over the displayed grid.

groups and the assignments randomly permuted. Conditional on treatment status x_i, the survival time is sampled from an exponential distribution, $T_i \sim \text{Exp}(\lambda e^{bx_i})$ for $i = 1, \cdots, n$, so that the baseline hazard is λ and the treatment hazard ratio is e^b.

Right-censoring is imposed in one of two ways. In the simulations using random censoring, the censoring time is sampled independently as $C_i \sim \text{Exp}(\lambda_c)$ independently of both the survival time and treatment assignment, corresponding to non-informative right censoring. Alternatively, when a maximum follow-up time is specified, all subjects are administratively censored at the fixed time $C_i = T_{\max}$. The observed follow-up time is $Y_i = \min(T_i, C_i)$, with event indicator $\delta_i = \mathbf{1}_{T_i \leq C_i}$. Thus, subjects are sampled independently, and the censoring mechanism is non-informative by construction.

We used the number of observed events, $n^* = \sum_{i=1}^{n} \delta_i$, as the effective sample size in the eJAB formula. The simulation code and accompanying description are provided in `coxphSim.Rmd` in the GitHub repository.

- *'In particular, we tested the harmonic mean of group sizes for item 1, as in JAB (Wagenmakers, 2022), but do not demonstrate it here': I assume this is intended to investigate unequal sample sizes in the two groups? Should this not then be half the harmonic mean?*

The two sample-size considerations were not intended to address unequal group sizes per se. In the simulations, we used unequal group sizes with a ratio of 1.2:1, and we considered two sample-size conventions for computing Bayes factors: (1) the total number of observations and (2) one-half of the harmonic mean of the two group sizes.

The former convention is commonly used in linear-regression formulations and in the BIC approximation. The latter is also the inverse-variance effective sample size, $1/n = 1/n_1 + 1/n_2$, and is used in the JAB and JZS Bayes factors (Morey and Rouder, 2026). In the `ttestSim.Rmd` supplementary file in the GitHub repository, we included both sample-size definitions when computing JAB and BIC.

The use of one-half of the harmonic mean arises from the standard error of the difference between two independent group means. Wagenmakers (2022, pp. 26–27) discusses this sample-size choice in detail. Later, on p. 34, the central component is the $\sqrt{n}$ term—or, more generally, the sample-size quantity that enters the standard error of the maximum likelihood estimate.

- *Section 3.1: 'For local alternatives ...': Local alternatives only appear here, but they should also be discussed much more broadly in the theory section, as they form the basis of Johnson's BFs based on test-statistics.*

  We agree that local alternatives deserve a more prominent treatment in the theoretical development. We have expanded the discussion accordingly, including their role in TSBF and their relationship to our regularity conditions. These points are addressed in detail in the first bullet point of our response above.

- *Section 3.2: 'We compared eJAB with test-statistic BFs': How did you choose the parameter alpha (in Johnson's notation)? Or did you simply use the 'MMLE under the alternative hypothesis'? If so, it is not surprising that 'the test-statistic BF is inherently bounded below by 1'; it is actually a minimum BF! The discrepancies between your BF and Johnson's proposal are easy to explain, you are comparing apples with oranges.*

  We did indeed use the MMLE under the alternative hypothesis. In the revised manuscript, we instead compare the methods across several values of the hyperparameter and avoid comparisons based on the MMLE. These points are also addressed in detail in the first bullet point of our response above.

- *Beginning of Section 4: 'Prior work has raised concerns about the strength of statistical evidence in clinical research. For example, Begley and Ellis (2012) reported replication for only 11% of landmark preclinical cancer findings.' It does not make much sense to quote a study on the reproducibility of preclinical cancer findings as an example of 'concerns about the strength of statistical evidence in clinical research'. Clinical and preclinical research are very different fields.*

  Thank you for pointing this out. We have removed the reference from the revised manuscript.

- *The description of the 'candidate type I errors' is very repetitive, as you are always using the phrase 'We flag as a candidate type I error the reported effect ...' In addition, the reader would like to see additional information on sample size n (and dimension q) of the different studies, also because the focus is on the Jeffreys–Lindley paradox, a phenomenon that depends intrinsically on sample size. I noted that the first study is a four-arm Phase III trial, but only one p-value is reported. Is this the p-value for the comparison of all four groups or the p-value for the pairwise comparison of 50 mg vs placebo? What sample size was used, and did you also analyze the results from the other pairwise comparisons with different doses? If so, how did you deal with the dependence between the comparisons, given that they all share the same control arm?*

  In the revised manuscript, we report effect size, sample size, trial phase, and the outcome and comparison for each example. The three examples have $n = 66$, 80, and 1,586, respectively, with $q = 1$ in each case. The original four-arm Phase III trial is excluded by the revised eligibility criteria.

More generally, a single study may contribute multiple p-values. For the posterior summaries, we aggregate results within studies into four categories defined by significance and whether eJAB exceeds 1. We use a Dirichlet$(1, 1, 1, 1)$ prior for each study and weight posterior category probabilities by the number of eligible analyses before calculating the contradiction proportion among significant results. By assigning each study its own category probabilities, the model allows for study-level heterogeneity.

- *End of Section 4: The linear model needs better motivation. Figure 4 would be more informative if the same y-axis scale were used throughout and if the panels were ordered A, C, B, D (for easier comparison of unadjusted and adjusted results).*

  Thank you for this suggestion. We have adjusted the y-axis scale to be consistent across all panels and reordered the panels A, C, B, D as suggested to facilitate comparison of the unadjusted and adjusted results.

  The linear mixed models describe within-study variation in $\ln eJAB$ and $\ln p$, separately for primary and secondary outcomes. We fit a fixed intercept and study-specific random intercept, and plot the residuals re-centered by the estimated fixed intercept.

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Response to Reviewer 2

There is no doubt that this work will attract a lot of attention. Similar to the 2018 Benjamin et al. paper, the approach outlined in this paper is straightforward to implement, and, in my opinion, there is a good argument to be made that when $p < \alpha$ ('reject $\mathcal{H}_0$') while $BF_{01} > 1$ ('the data support $\mathcal{H}_0$'), or even < 1 but $> 1/3$ ('the data provide weak evidence against $\mathcal{H}_0$'), this casts substantial doubt on the frequentist conclusion – at any rate, such a conflict demands a more careful analysis. The fact that such conflicts now go entirely unnoticed seems unproductive. It may be that, at several points throughout the paper, its accessibility can be increased even more. Regardless, the work provides a fresh perspective on a crucial issue that continues to haunt the field. It is certain to lead to a useful debate, and, in fact, I would suggest that –if it is accepted– it will be the target of a few comments.

Thank you very much for this encouraging and thoughtful assessment. We appreciate the reviewer's recognition of the practical relevance and potential impact of the proposed approach. We also agree that conflicts between NHST and Bayes-factor conclusions warrant closer scrutiny rather than going unnoticed (e.g., Aust et al., 2026; Tyner et al., 2025).

In the revised manuscript, we have expanded the theoretical content to provide a stronger justification for the methodology. We also agree on the importance of making the methodology more accessible. To this end, we have developed an eJAB calculator, available at `https://flxp0x-puneet-velidi.shinyapps.io/bayesfactor/`, and implemented eJAB in a JASP module designed to identify potentially irreplicable results. A companion article describes the JASP module and presents three examples in which it is used to identify, from a collection of studies, those that are unlikely to replicate when replication outcomes are subsequently available.

The manuscript's Equally Weighted Decision Error Profiles compare eJAB with the familiar NHST significance rule $p < .05$, TSBFs evaluated at true and estimated effects, and BIC approximations. These profiles show where each rule has lower decision error across sample sizes, effect sizes, and dimensions.

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